

In-Person Lab - Disassemble a Desktop Computer

Version 1.0 November 2023

Before You Begin - How to Complete This Activity

Please refer to the [Lab Process Guide](#) for instructions on completing and submitting this activity.

Introduction

In this lab, you will experience disassembling a desktop computer. You will also recognize and embed images of various hardware components into this lab file.

Safety first!

To ensure your safety and the safety of the electronic components, always verify that the computer is disconnected from any power source, and use the antistatic gear when disassembling it.

Note:

Different computer cases are assembled in different ways. In this lab, you will follow the steps for disassembling a desktop computer in a generally logical manner. However, you may still encounter situations where following the prescribed steps in sequential order is impossible. One hardware component may get in the way of removing another component. If this occurs, use your common sense and jump to the relevant step non-sequentially.

Objectives

At the end of this lab, you will be able to disassemble a desktop computer.

Equipment/Requirements

A standard desktop computer (one per learner) in working condition that includes the following:

- Computer case.
- Power supply unit.
- CPU.
- Motherboard with live CMOS battery.

- Graphic card.
- RAM.
- Case fan.
- CPU cooler.
- HDD.
- SSD M.2.
- DVD (if available).
- PCI NIC.
- Internal cables.
- Monitor, keyboard, and mouse.
- Power cable.

In addition to the above list:

- Learner laptops with internet connection.
- Power outlet.
- A cellphone with a camera.
- Tech toolkit for each learner.

Disassemble a desktop computer.

Step 1: Verify that the computer works.

- Set up your working environment. Ensure that there is adequate lighting and antistatic protection.*
- Place the desktop computer on the workbench. Label the computer with your name.*
- Connect the monitor, keyboard, and mouse, and plug the computer into a power outlet.*
- Power on the computer. If your computer does not power on, notify the instructor.*
- Did the computer display the BIOS splash screen and BIOS access keys? Enter your answer into the box below.*

Yes

- Did the computer load an operating system? Enter your answer into the box below.*

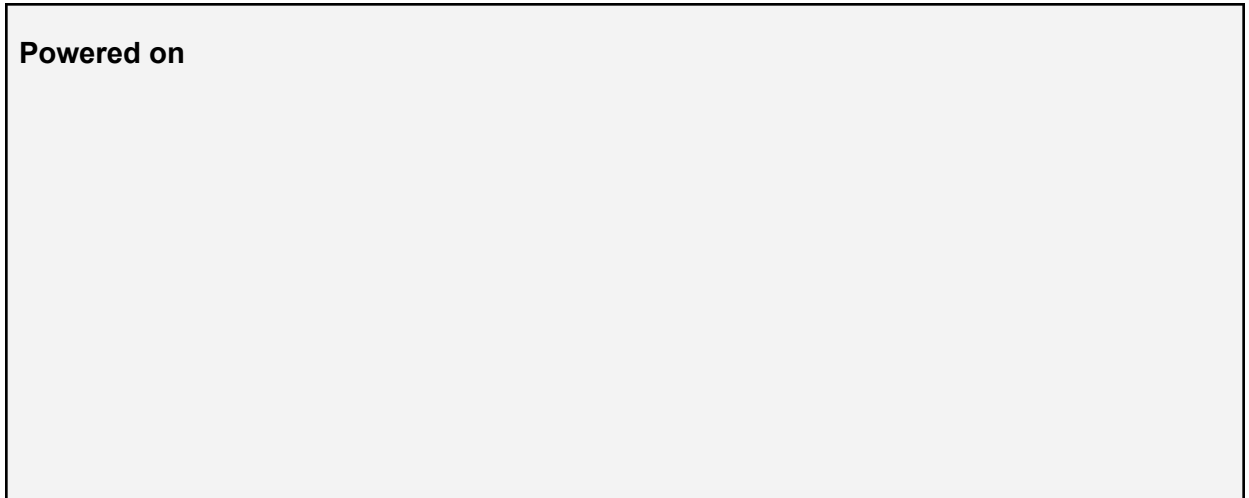
Yes

- Type the make and model of your computer into the box below (if one exists).*

Make: Dell Model: Opti Plex 9020 SFF

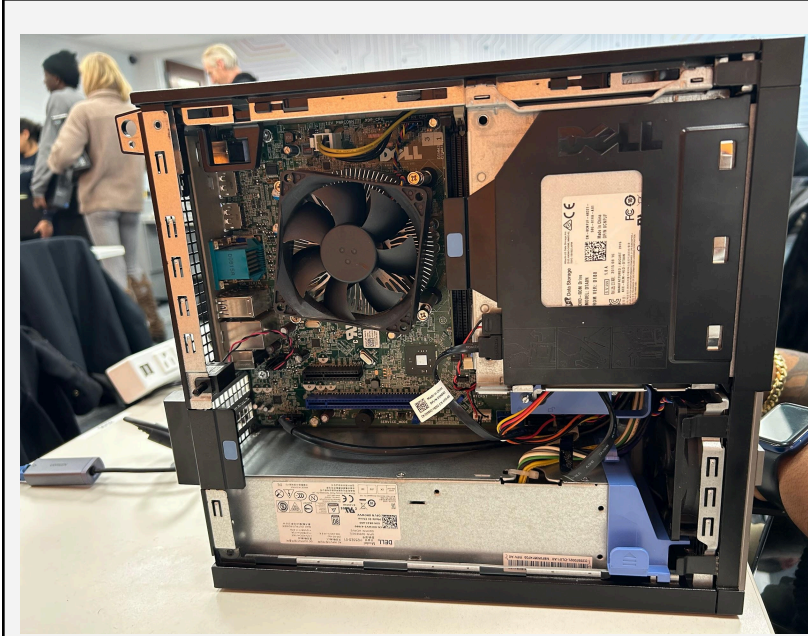
- h. *Use your smartphone to take a picture of the powered-on computer. Place the image in the box below.*

Powered on



Step 2: Open the case.

- a. ***Use the antistatic wrist strap to protect components from static electricity discharge (ESD). If you do not have a properly grounded antistatic wrist strap, you can touch a **conductive metallic** object for 10 seconds before you begin work.***
- b. *Identify the location of the **screws** holding the side panels to the case frame.*
Note: Not all cases have screws or require a screwdriver to remove the side panels.
- c. *Release the screws holding the side panels to the case frame. Use a screwdriver if necessary. Make sure you put the screws in a **marked container**, as computer cases have different screws for securing different hardware devices.*
- d. *Remove the side panels from the case.*
- e. *Using your smartphone, take a photo of your **open computer case** and place the image in the box below.*



Step 3: Disconnect all internal cables.

There are two main types of internal cables in a computer case: 1) Power cables, and 2) Data/control cables. In this step, you will identify and disconnect the various internal cables.

Be sure to take a photo of the cables connecting the case's front panel to the motherboard **BEFORE** disconnecting them to make restoring this connection simple.

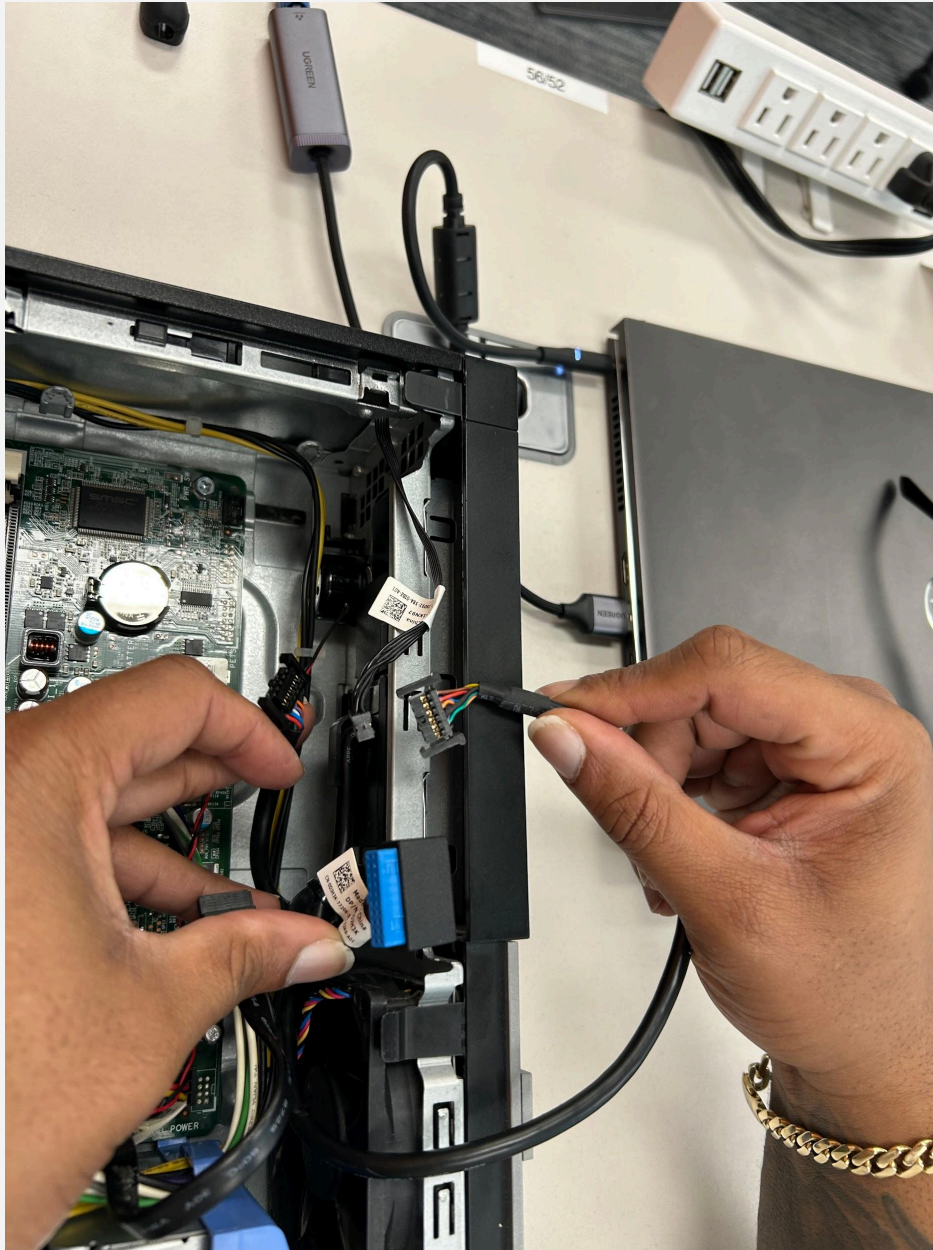
- a. *Start disconnecting internal cables following the order described in the next steps (if possible). Note that some cables may have a clip holding them in place.*
- b. *Disconnect the main power cable. Once you have disconnected the main power cable connecting the motherboard, use your smartphone to take a picture of the disconnected cable connector and place the image in the box below.*



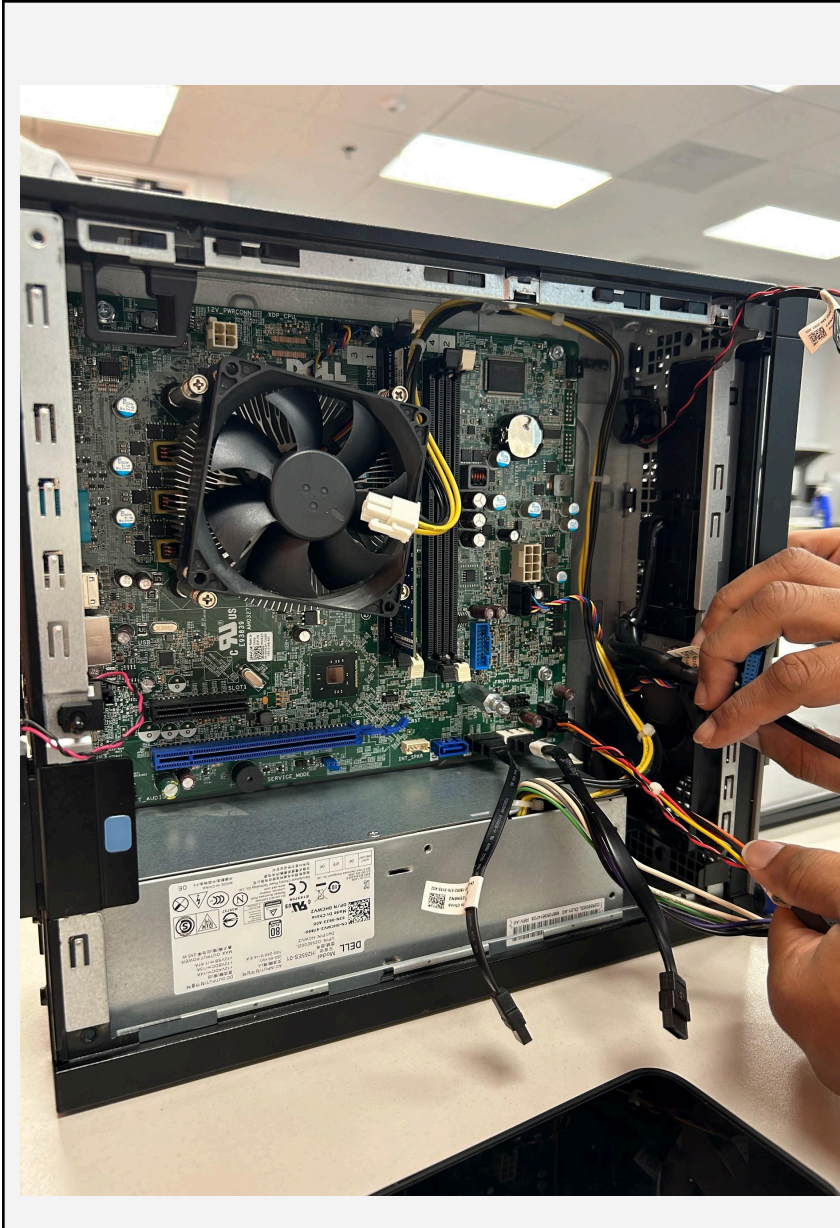
- c. *Continue disconnecting cables. Once you have disconnected a SATA data cable, use your smartphone to take a picture of the disconnected cable and place the image in the box below.*



- d. *Identify the front panel cables connecting the case's front panel, LEDs, and power switch to the motherboard. Use your smartphone to take a picture of the cables connecting the motherboard **BEFORE** you disconnect them. Place the image in the box below.*



- e. *Disconnect the front panel cables.*
- f. *Finish disconnecting all remaining internal cables.*
- g. *Using your smartphone, take a photo of your open computer case after all of the cables have been disconnected and removed, and place the image in the box below.*

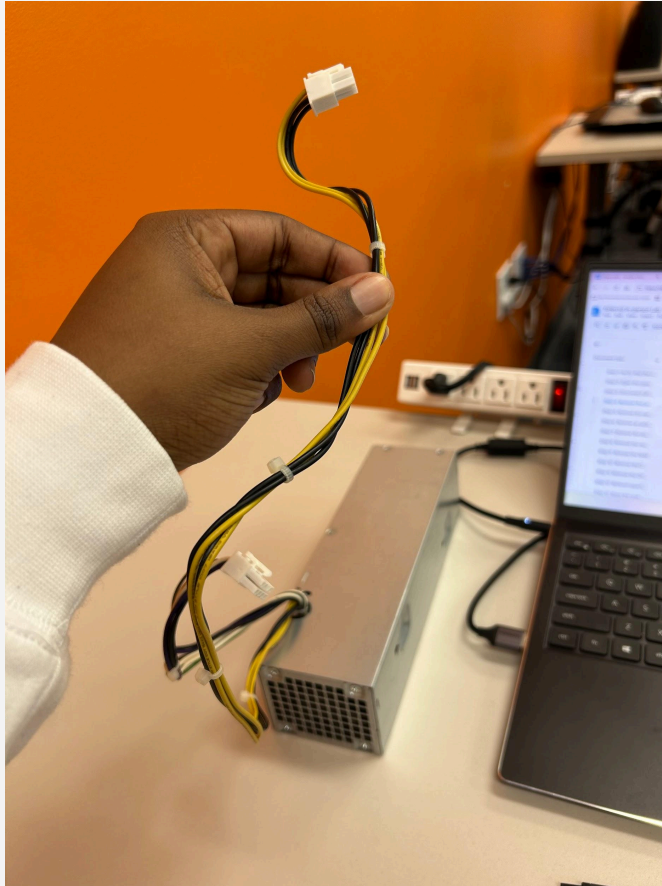


- h. *Put all loose cables in a labeled container, identifying the cables as belonging to the computer you are working on.*

Step 4: Remove the power supply unit.

- a. *Identify the location of the **screws** that secure the power supply unit to the case.*
Note: Not all cases have screws or require a screwdriver to remove power supplies; others may have a panel or cover holding the power supply in place.
- b. *Release the screws securing the power supply. Use a screwdriver if necessary. Make sure that you put the screws in a **marked container**, as computer cases have different screws for securing different hardware devices.*
- c. *Remove the power supply from the computer case and place it in your work area.*

- d. Using your smartphone, take a photo of the power supply and place the image in the box below.



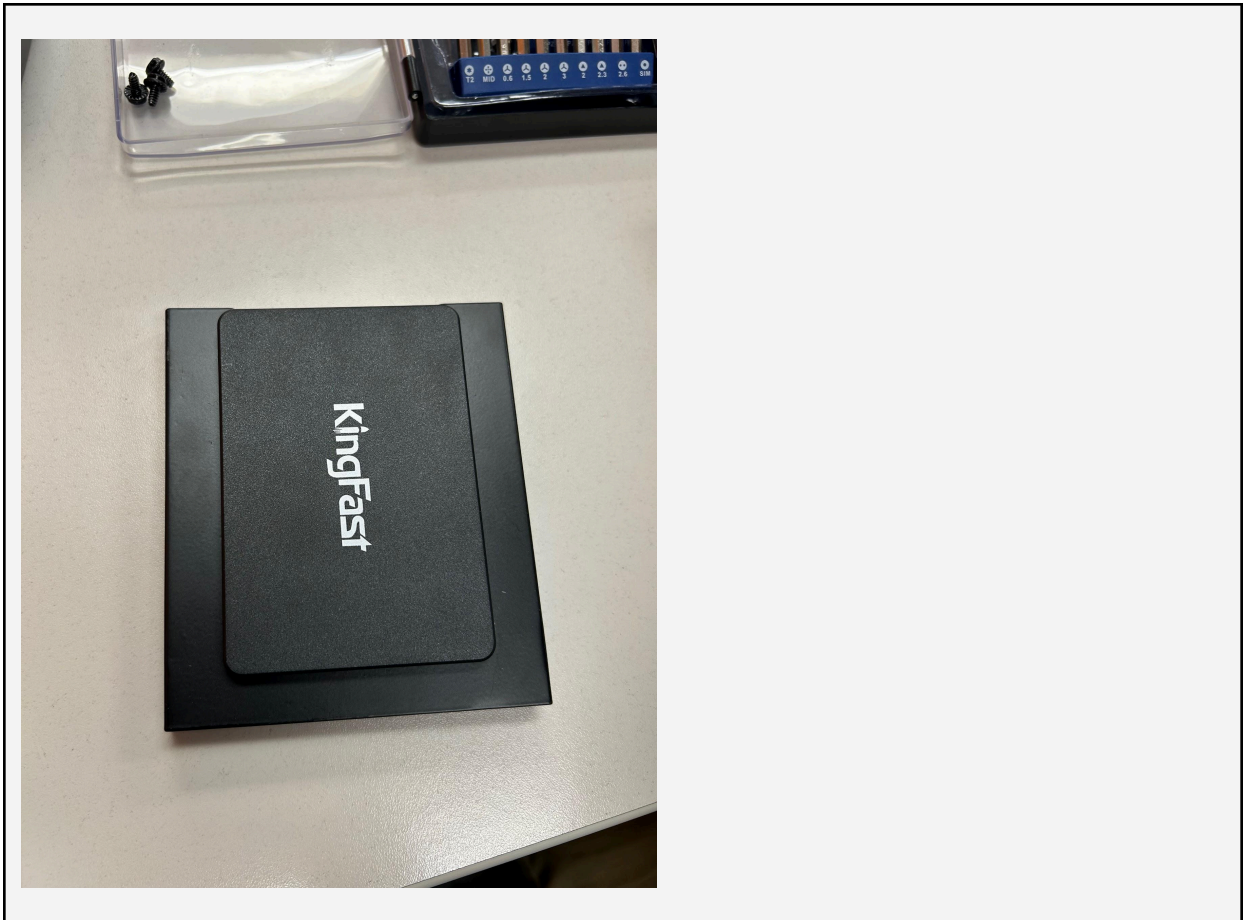
Step 5: Remove the network adapter card.

- a. Does your computer have a network adapter card installed into an expansion slot? Type your answer into the box below.

No

- b. If the answer is "**No**," move on to Step 6; otherwise, continue in this step.
- c. Identify the location of the **screws** securing the network adapter card to the case.
Note: Not all cases have screws or require a screwdriver to remove extension cards; others may have a panel holding all cards in place.
- d. Release the screws securing the network adapter card to the case. Use a screwdriver if necessary. Make sure you place the screws in a **marked container**, as computer cases have different screws for securing different hardware devices.
- e. Release the clip securing the card to the extension slot.
- f. Remove the card from the case and lay it on a **non-conductive surface**. **Do not touch the golden conductors on the card.**

- g. Using your smartphone, take a photo of the network adapter card and place the image in the box below.



- h. Place the network adapter card in an antistatic bag (if you have one).
i. Label the antistatic bag containing the network adapter card as belonging to your computer.

Step 6: Remove an additional extension card.

- a. Does your computer have another expansion card(s) other than a graphics card or a network adapter installed into an expansion slot? Type your answer into the box below.

No

- b. If the answer is “No,” move on to Step 7; otherwise, continue in this step.
c. Identify the location of the **screws** securing the expansion card to the case.
Note: Not all cases have screws or require a screwdriver to remove extension cards; others may have a panel holding all cards in place.
d. Release the screws securing the expansion card to the case. Use a screwdriver if necessary. Make sure you place the screws in a **marked container**, as computer cases have different screws for securing different hardware devices.
e. If applicable, release the clip securing the card to the extension slot.

- f. Remove the card from the case and lay it on a **non-conductive surface**. **Do not touch the golden conductors on the card.**
- g. Repeat this process for any additional expansion card.
- h. Using your smartphone, take a photo of the expansion card(s) and place the image in the box below.

N/A

- i. Place the expansion card(s) in an antistatic bag (if you have one).
- j. Label the antistatic bag containing the expansion card as belonging to your computer.

Step 7: Remove the storage device(s).

In this step, you will remove all internal storage devices from the computer. These may include one or more HDDs, SSDs, and optical drives. Start by answering the following questions to determine the computer's storage setup and move forward with removing and researching the devices.

Note that different computer cases offer different bays to accommodate storage devices, and bays may be located in different locations. In addition, storage devices using interfaces such as M.2 may be connected directly to the motherboard and hidden under a cooling plate. You can find the storage bay's location in the case by consulting the case manual if necessary. The case manual is usually found on the manufacturer's website.

Regardless of their number, you will remove all the storage devices from the case.

- a. Explore and identify all the storage devices. How many storage devices did you find? Type your answer into the box below.

1

- b. Identify the location of the **screws** securing the storage device in the case.
Note: Not all cases have screws or require a screwdriver to remove the storage device.

- c. Release the screws securing the storage device to the bay. Use a screwdriver if necessary. Make sure you place the screws in a **marked container**, as computer cases have different screws for securing different hardware devices.
- d. Remove the storage device from the computer case and place it in your work area. Repeat substeps b - d for any additional storage devices.
- e. Using your smartphone, take a photo of the storage device(s) and place the image in the box below.



- f. Place the storage device(s) in an antistatic bag (if you have one).
- g. Label the antistatic bag containing the storage device(s) as belonging to your computer.

Step 8: Remove the motherboard.

Usually, having the motherboard on the workbench at this point allows for better access to the RAM and CPU. This is the reason we are taking this step now. Still, there may be cases where removing elements such as the CPU cooler are necessary to allow access to the motherboard screws. Determine the situation and make a decision as to how you wish to proceed. If you do

not have access to the motherboard's screws, or if the case fans do not allow the removal of the motherboard, skip this step, remove some other components, and return to this step later.

- a. *Identify the location of the **screws** securing the motherboard to the case.*
Note: Some cases have the motherboard mounted to a detachable plate.
- b. *Release the screws that secure the motherboard. Use a screwdriver if necessary. Make sure you put the screws in a **marked container**, as computer cases have different screws for securing different hardware devices.*
- c. *Gently remove the motherboard from the computer case. Be careful not to damage the I/O plate on the case encasing the motherboard's ports. **Avoid touching the conductive parts** on the motherboard. If the CPU cooler is present, you can use it to lift the motherboard out of the case. Place the motherboard in your work area on a **non-conductive surface**.*

- d. Using your smartphone, take a photo of the motherboard and place the image in the box below.



- e. **Only** if you have **already removed** the RAM, CPU cooler, and CPU from the motherboard, place the motherboard in an antistatic bag (if you have one).
f. Label the antistatic bag containing the motherboard as belonging to your computer.

Step 9: Remove the RAM.

- a. Explore the motherboard and identify the location of the RAM modules

- b. How many RAM modules are installed on the motherboard? Type the answer in the box below.

1

- c. Regardless of the number of RAM modules, release the clips holding the RAM in place and remove the RAM from the motherboard. **Do not touch the gold-plated conductive area.**
- d. Place the RAM modules on a **non-conductive surface**. Using your smartphone, take a photo of the RAM modules and place the image in the box below.



- e. Place the RAM module(s) in an antistatic bag (if you have one).
- f. Label the antistatic bag containing the RAM module(s) as belonging to your computer.

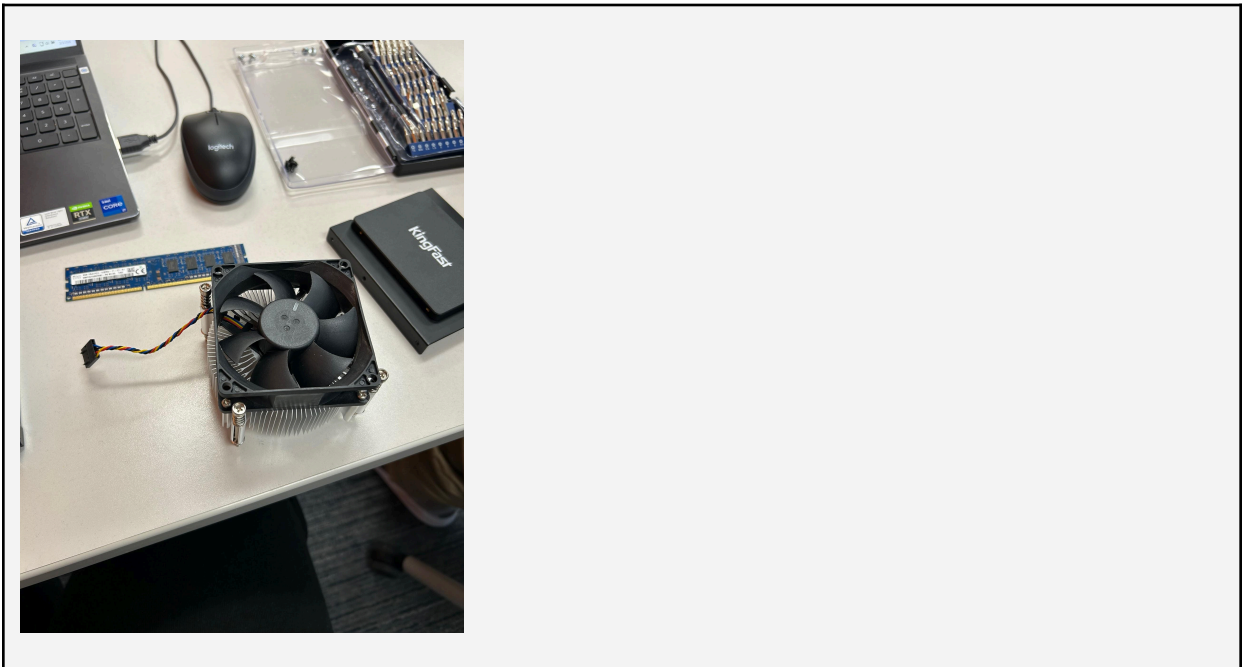
Step 10: Remove the CPU cooling.

With CPU cooling being an extremely extensive world, removing the CPU cooling can be very easy or seriously challenging. If you run into problems with removing the CPU cooling, conduct some research over the internet as to how the cooler is installed. There are just no end-to-cooling options for CPUs.

- a. Identify the mechanism securing the CPU cooler to the motherboard.

Note: Not all CPU cooling units require a screwdriver for removal; others may have thumb screws, levers, or both holding the cooler in place.

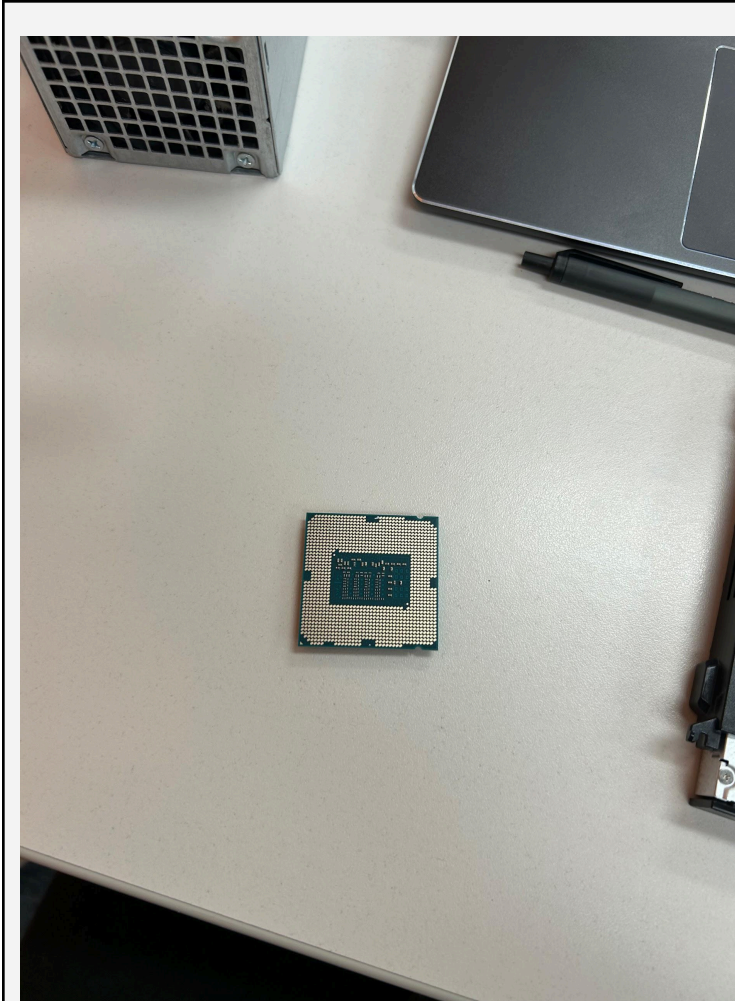
- b. *Ensure that the CPU cooler power cable is disconnected. Release the mechanism securing the cooler to the CPU. Use a screwdriver if necessary. Make sure you put the screws and other parts in a **marked container**.*
- c. *Once the CPU cooler is loose, gently remove it from the CPU. Note that old thermal paste may cause the cooler to stick to the CPU. Take care not to yank the CPU out of its socket when you remove the cooler.*
- d. *Use your smartphone to take a picture of the CPU cooler. Place the image in the box below.*



- e. *Place the CPU cooler into a container (if you have one).*
- f. *Label the container with the CPU cooler as belonging to your computer.*

Step 11: Remove the CPU.

- a. *Identify the location of the lever securing the CPU to the socket.*
Note: Different CPU sockets have different mechanisms to secure the CPU. Regardless, all will have some type of lever you need to release. Some sockets will have an additional cover plate you need to open. All CPU sockets are ZIF; therefore, do not use force.
- b. *Release the mechanism securing the CPU to the socket. Usually, no tools are necessary.*
- c. *Gently lift the CPU out of the socket. You should lift the CPU up directly without applying force. **Do not touch the CPU's conductive area or pins**, and place it in your work area on a **non-conductive surface, pins/contacts facing up**.*
- d. *Use your smartphone to take a picture of the CPU. Place the image in the box below.*



- e. *Place the CPU in a protective container (if you have one).*
- f. *Label the container with the CPU as belonging to your computer.*
- g. *If you have not yet done so, place the motherboard into an antistatic bag.*
- h. *Label the antistatic bag containing the motherboard as belonging to your computer.*

Step 12: Remove case fans.

As with other components, the desktop computer's case fans come in various numbers and sizes. Some have LED lights, others do not. Some are quiet, others are loud, some come with speed control, others do not, and so on.

Regardless, desktop case fans are critical in keeping the computer cool and running smoothly. In this step, you will document the number and location of the case fans and then remove the fans from the case.

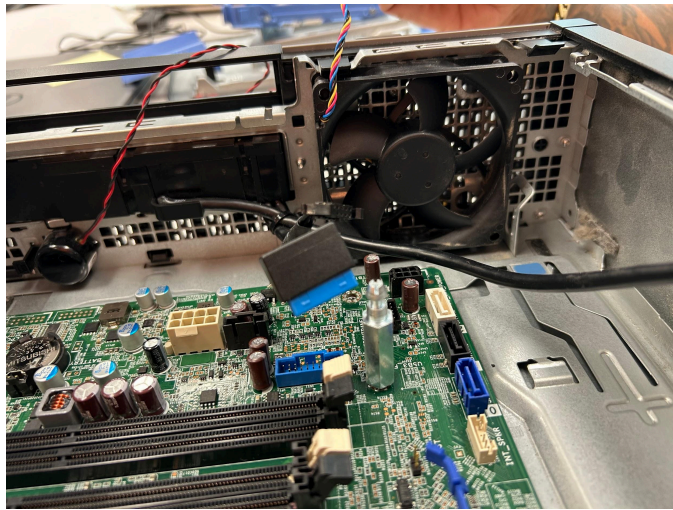
- ☐ *Explore the desktop case and identify the location of all the case fans.*

Note: Fans may be tucked behind the front panel and on the top/bottom of the case.

- ☐ How many case fans did you find attached to your case? Type the answer in the box below.

1

- ☐ Document and label the location of the fans before you remove them so that when the time comes, you can mount them back to the original location.
- ☐ Remove all of the case fans and lay them on the workbench.
- ☐ Place all of the fan screws in a labeled container.
- ☐ Use your smartphone to take a picture of the case fans. Place the image in the box



below.

Place the fans into a container.

- ☐ Label the container with the fans as belonging to your computer.

Step 13: Store the computer and clear the workbench (optional).

This step is optional based on the plans for what is coming next. If the plan is to perform the computer assembly lab immediately, there is no need to store the dismantled computer, so you can discard this step; otherwise, proceed with the following steps.

- Gather up all the desktop computer components and move them to the storage area.
- Gather all of the tools and antistatic gear and move them to the storage area.
- Ensure that your work area is tidy and clean.